

Output tables for 1xN statistical comparisons.

March 10, 2025

1 Average rankings of Friedman test

Average ranks obtained by each method in the Friedman test.

Algorithm	Ranking
Random	6.875
NN	5.3125
Sweep	5.8125
SaveSeq	1.75
SaveParall	2.375
MJ	2
CMT	3.875

Table 1: Average Rankings of the algorithms (Friedman)

Friedman statistic (distributed according to chi-square with 6 degrees of freedom): 42.84375.
P-value computed by Friedman Test: 0.

2 Post hoc comparison (Friedman)

P-values obtained in by applying post hoc methods over the results of Friedman procedure.

i	algorithm	$z = (R_0 - R_i)/SE$	p	Holm	Finner	Li
6	Random	4.744828	0.000002	0.008333	0.008512	0.009634
5	Sweep	3.761144	0.000169	0.01	0.016952	0.009634
4	NN	3.298234	0.000973	0.0125	0.025321	0.009634
3	CMT	1.967368	0.049141	0.016667	0.033617	0.009634
2	SaveParall	0.578638	0.562834	0.025	0.041844	0.009634
1	MJ	0.231455	0.816961	0.05	0.05	0.05

Table 2: Post Hoc comparison Table for $\alpha = 0.05$ (FRIEDMAN)

Holm's procedure rejects those hypotheses that have an unadjusted p-value ≤ 0.016667 .
Finner's procedure rejects those hypotheses that have an unadjusted p-value ≤ 0.033617 .
Li's procedure rejects those hypotheses that have an unadjusted p-value ≤ 0.009634 .

3 Adjusted P-Values (Friedman)

Adjusted P-values obtained through the application of the post hoc methods (Friedman).

i	algorithm	unadjusted p	p_{Holm}
1	Random	0.000002	0.000013
2	Sweep	0.000169	0.000846
3	NN	0.000973	0.003892
4	CMT	0.049141	0.147423
5	SaveParall	0.562834	1.125668
6	MJ	0.816961	1.125668

Table 3: Adjusted p -values (FRIEDMAN) (I)

i	algorithm	unadjusted p	p_{Finner}	p_{Li}
1	Random	0.000002	0.000013	0.000011
2	Sweep	0.000169	0.000507	0.000923
3	NN	0.000973	0.001945	0.005287
4	CMT	0.049141	0.072798	0.21165
5	SaveParall	0.562834	0.62951	0.754598
6	MJ	0.816961	0.816961	0.816961

Table 4: Adjusted p -values (FRIEDMAN) (II)